Introduction to Statistics - Quiz #4(60 minutes)

June 4, 2026 (Thursday)

Section(교반): _____ Cadet Number(교번): _____ Name(성명): _____ Score: _____

- All solutions must include a detailed step-by-step explanation.
- If an answer has more than four decimal places, round to the **fourth decimal place**.
- **Reference table** is provided on the last page of the exam.

1. The driving distance per full charge (in km) of an electric vehicle is known to follow a normal distribution: with a new battery (A), mean μ_A and variance $\sigma_A^2 = 18$; with the existing battery (B), mean μ_B and variance $\sigma_B^2 = 22$. To test whether the average distance differs between the two batteries, $n_A = 9$ vehicles using battery (A) and $n_B = 11$ vehicles using battery (B) were randomly selected, giving distances X_{Ai} $(i = 1, \ldots, 9)$ and X_{Bj} $(j = 1, \ldots, 11)$. Conduct a hypothesis test on the difference of two means at significance level $\alpha = 0.05$. [55 points]

(1) State the null and alternative hypotheses.

Solution:

$$H_0 : \mu_A - \mu_B = 0, \qquad H_A : \mu_A - \mu_B \neq 0$$

(2) Find the null distribution of the test statistic

$$Z = \frac{\bar{X}_A - \bar{X}_B - 0}{\sqrt{\dfrac{\sigma_A^2}{n_A} + \dfrac{\sigma_B^2}{n_B}}},$$

where $\bar{X}_A$ is the sample mean for the new battery and $\bar{X}_B$ is the sample mean for the existing battery.

Solution: Both population variances are known and $\bar{X}_A - \bar{X}_B \sim N(\mu_A - \mu_B, \sigma_A^2/n_A + \sigma_B^2/n_B)$, so under H_0,

$$Z = \frac{\bar{X}_A - \bar{X}_B - 0}{\sqrt{\frac{\sigma_A^2}{n_A} + \frac{\sigma_B^2}{n_B}}} = \frac{\bar{X}_A - \bar{X}_B}{\sqrt{4}} \sim N(0,\, 1).$$

(3) Find the rejection region at significance level $\alpha = 0.05$.

Solution: This is a two-sided test with $\alpha = 0.05$, so the rejection region is

$$|z| > z_{0.025} = 1.96.$$

(4) The following summary statistics were obtained. Find the observed value of the test statistic in (2).

$$\bar{x}_A = \frac{1}{9}\sum_{i=1}^{9} x_{Ai} = 455, \qquad \bar{x}_B = \frac{1}{11}\sum_{j=1}^{11} x_{Bj} = 449.5455$$

Solution: Since $\bar{x}_A = 455$ and $\bar{x}_B = 449.5455$,

$$z = \frac{455 - 449.5455}{\sqrt{\frac{18}{9} + \frac{22}{11}}} = 2.7273.$$

(5) Using the results in (3) and (4), complete the hypothesis test. State the conclusion in the context of the data.

Solution: The observed statistic $z = 2.7273$ lies in the rejection region $|z| > 1.96$, so we reject H_0.

Conclusion: At significance level $\alpha = 0.05$, there is sufficient evidence to conclude that the average distance of the new battery differs from that of the existing battery.

2. We wish to investigate whether there is a positive linear relationship between the summer daily maximum temperature X and a café's delivery sales Y. To do so, $n = 31$ days were randomly sampled, and for each day the maximum temperature and the café's delivery sales were recorded, giving the results below. Assume that (X, Y) follows a bivariate normal distribution. [45 points]

i	1	2	3	4	5	6	7	$\cdots$	30	31
Max. temperature (x_i, °C)	33.2	32.5	34.2	36.2	33.7	33.5	33.4	$\cdots$	32.9	33
Delivery sales (y_i, 10,000 KRW)	30	25	40	55	20	27	22	$\cdots$	30	40

(1) State the null and alternative hypotheses.

Solution:

$$H_0 : \rho = 0, \qquad H_A : \rho > 0$$

(2) Find the null distribution of the test statistic

$$T = \frac{R\sqrt{n-2}}{\sqrt{1-R^2}},$$

where R is the sample correlation coefficient.

Solution: With $n = 31$, under H_0,

$$T = \frac{R\sqrt{n-2}}{\sqrt{1-R^2}} = \frac{R\sqrt{29}}{\sqrt{1-R^2}} \sim t(29).$$

(3) Find the rejection region at significance level $\alpha = 0.05$.

Solution: This is a right-sided test, so the rejection region is $t > t_\alpha(n-2)$. With $\alpha = 0.05$ and $n = 31$,

$$t > t_{0.05}(29) = 1.6991.$$

(4) The R code and its partial output for this problem are shown below. Use them to find the p-value and complete the hypothesis test at significance level $\alpha = 0.05$. State the conclusion in the context of the data.

```
> temp <- c(33.2, 32.5, 34.2, 36.2, 33.7, 33.5, 33.4, ..., 32.9, 33)
> sales <- c(30, 25, 40, 55, 20, 27, 22, ..., 30, 40)
> cor.test(temp, sales, alternative = 'greater')
```

```
        Pearson's product-moment correlation

data:  temp and sales
t = 4.0776, df = _______ , p-value = _________
alternative hypothesis: true correlation is greater than 0
95 percent confidence interval:
 0.3696727 1.0000000
sample estimates:
     cor
0.603664
```

Solution: The observed test statistic is $t = 4.0776$ with $df = n - 2 = 29$. For a right-sided test,

$$p\text{-value} = P(T > 4.0776) = 0.0002.$$

Since the p-value is smaller than $\alpha = 0.05$, we reject H_0.

Conclusion: At significance level $\alpha = 0.05$, there is sufficient evidence of a positive correlation between the summer daily maximum temperature and the café's delivery sales.

Reference Table

$\psi = \dfrac{\left(s_1^2/n_1 + s_2^2/n_2\right)^2}{\left(s_1^2/n_1\right)^2/(n_1-1) + \left(s_2^2/n_2\right)^2/(n_2-1)}, \qquad S_p^2 = \dfrac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}$			
$\Phi(0.005) = 0.502$	$\Phi(0.01) = 0.504$	$\Phi(0.025) = 0.51$	$\Phi(0.05) = 0.5199$
$\Phi(0.5) = 0.6915$	$\Phi(1) = 0.8414$	$\Phi(1.0607) = 0.8556$	$\Phi(1.3416) = 0.9101$
$\Phi(1.5) = 0.9332$	$\Phi(2) = 0.9772$	$\Phi(3) = 0.9987$	$\Phi(2.451) = 0.9929$
$z_{0.005} = 2.5759$	$z_{0.01} = 2.3263$	$z_{0.0228} = 1.999$	$z_{0.025} = 1.96$
$z_{0.0328} = 1.8417$	$z_{0.05} = 1.6449$	$z_{0.1} = 1.2816$	$z_{0.0668} = 1.5$
$t_{0.0085}(8) = 3$	$t_{0.0185}(8) = 2.5$	$t_{0.0403}(8) = 2$	$t_{0.1733}(8) = 1$
$t_{0.1599}(8) = 1.0607$	$t_{0.6633}(8) = -0.4375$	$t_{0.5562}(8) = -0.1458$	$t_{0.5207}(8) = -0.0536$
$t_{0.005}(8) = 3.3554$	$t_{0.01}(8) = 2.8965$	$t_{0.025}(8) = 2.3060$	$t_{0.05}(8) = 1.8595$
$t_{0.0075}(9) = 3$	$t_{0.0169}(9) = 2.5$	$t_{0.0383}(9) = 2$	$t_{0.1717}(9) = 1$
$t_{0.1582}(9) = 1.0607$	$t_{0.6640}(9) = -0.4375$	$t_{0.5564}(9) = -0.1458$	$t_{0.5208}(9) = -0.0536$
$t_{0.005}(9) = 3.2498$	$t_{0.01}(9) = 2.8214$	$t_{0.025}(9) = 2.2622$	$t_{0.05}(9) = 1.8331$
$t_{0.9998}(27) = -4.0776$	$t_{0.9998}(28) = -4.0776$	$t_{0.9998}(29) = -4.0776$	$t_{0.9998}(30) = -4.0776$
$t_{0.005}(27) = 2.7707$	$t_{0.01}(27) = 2.4727$	$t_{0.025}(27) = 2.0518$	$t_{0.05}(27) = 1.7033$
$t_{0.005}(28) = 2.7633$	$t_{0.01}(28) = 2.4671$	$t_{0.025}(28) = 2.0484$	$t_{0.05}(28) = 1.7011$
$t_{0.005}(29) = 2.7564$	$t_{0.01}(29) = 2.4620$	$t_{0.025}(29) = 2.0452$	$t_{0.05}(29) = 1.6991$
$t_{0.005}(30) = 2.75$	$t_{0.01}(30) = 2.4573$	$t_{0.025}(30) = 2.0423$	$t_{0.05}(30) = 1.6973$